

Technical University of Denmark

Advanced machine learning (02460)

Practice Exam Set C

Exam duration: 2 hours. Aid: None.

IMPORTANT: Questions are self-contained; all necessary formulas are provided.

Only information written on pages 2–11 will be evaluated.

Write your student number on all pages at the bottom-left.

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Part I Generative models

Exercise A — Probabilistic PCA

Consider a probabilistic PCA model with latent variable $z \in \mathbb{R}$ and observed variable $\mathbf{x} \in \mathbb{R}^2$. Let z follow a standard Gaussian prior $p(z) = \mathcal{N}(z \mid 0, 1)$, and the likelihood be

$$p(\mathbf{x} \mid z) = \mathcal{N}(\mathbf{x} \mid [2, 1]^\top z + [0, 1]^\top, \mathbf{I}_2).$$

Recall that if $z \sim \mathcal{N}(0, 1)$ and $\mathbf{x} = \mathbf{w}z + \mathbf{b} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, then

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \mathbf{b}, \mathbf{w}\mathbf{w}^\top + \mathbf{I}). \quad (1)$$

Question A.1: What is the marginal distribution $p(\mathbf{x})$?

- A ☐ $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid [0, 1]^\top, \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix})$
- B ☐ $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid [0, 1]^\top, \begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix})$
- C ☐ $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid [0, 0]^\top, \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix})$
- D ☐ $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid [0, 0]^\top, \begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix})$

Give a brief argument for your answer.

Exercise B — Normalizing flows

Consider a normalizing flow with $\mathbf{u}, \mathbf{x} \in \mathbb{R}^2$, where $p_{\mathbf{u}}(\mathbf{u}) = \mathcal{N}(\mathbf{u} \mid \mathbf{0}, \mathbf{I}_2)$ and the transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is

$$T(\mathbf{u}) = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{u} + [1, -1]^\top. \quad (2)$$

The density of $\mathbf{x} = T(\mathbf{u})$ is

$$p_{\mathbf{x}}(\mathbf{x}) = p_{\mathbf{u}}(T^{-1}(\mathbf{x})) \cdot |\det J_{T^{-1}}(\mathbf{x})|. \quad (3)$$

Question B.1: What is $p_{\mathbf{x}}([1, -1]^\top)$?

- A ☐ $\frac{1}{6\pi}$
- B ☐ $\frac{1}{12\pi}$
- C ☐ $\frac{1}{18\pi}$
- D ☐ $\frac{1}{36\pi}$

Give a brief argument for your answer.

Exercise C — Denoising Diffusion Probabilistic Models

Consider a DDPM with $T = 5$ steps. The variance schedule is

$$\beta_1 = \frac{1}{4}, \quad \beta_2 = \frac{1}{3}, \quad \beta_3 = \frac{1}{2}, \quad \beta_4 = \frac{1}{4}, \quad \beta_5 = \frac{1}{4}.$$

Recall the forward process in eq. (3), the marginal distribution

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t \mid \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad \bar{\alpha}_t = \prod_{\tau=1}^t (1 - \beta_\tau), \quad (4)$$

and the *forward posterior*

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1} \mid \tilde{\mu}_t, \tilde{\beta}_t \mathbf{I}), \quad (5)$$

where

$$\tilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 + \frac{\sqrt{1 - \beta_t} (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t, \quad \tilde{\beta}_t = \frac{\beta_t (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}. \quad (6)$$

Question C.1: Assuming $\mathbf{x}_0 = [0, 0]^\top$ and $\mathbf{x}_2 = [3, 0]^\top$, what is $q(\mathbf{x}_1 \mid \mathbf{x}_2, \mathbf{x}_0)$?

First compute $\bar{\alpha}_1$ and $\bar{\alpha}_2$. Then use equations (5) and (6).

$$\bar{\alpha}_1 = \quad \bar{\alpha}_2 =$$

$$q(\mathbf{x}_1 \mid \mathbf{x}_2 = [3, 0]^\top, \mathbf{x}_0 = [0, 0]^\top) =$$

Sketch your derivations below.

Part II Representation learning

Exercise D — Pull-back metrics

Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

$$f(\mathbf{x}) = \begin{bmatrix} x_1^2 + x_2, & x_1 + x_2^2, & x_1 x_2 \end{bmatrix}^\top. \quad (7)$$

Question D.1: Compute the Jacobian $J_{\mathbf{x}}$ of f at $\mathbf{x}$, and then derive the pull-back metric $G_{\mathbf{x}} = J_{\mathbf{x}}^\top J_{\mathbf{x}} \in \mathbb{R}^{2 \times 2}$.

Question D.2: Evaluate $G_{\mathbf{x}}$ at $\mathbf{x} = [1, 0]^\top$.

$$G_{[1,0]^\top} = \begin{bmatrix} & \\ & \end{bmatrix}$$

Exercise E — Geodesic approximation

Consider a generative model $\mathbf{y} = f(\mathbf{x})$, where $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{y} \in \mathbb{R}^D$ with $d < D$, and $f : \mathbb{R}^d \rightarrow \mathbb{R}^D$ is smooth. We wish to find the *geodesic* (shortest path on the manifold) between $\mathbf{c}_0$ and $\mathbf{c}_{N+1}$ by minimising over intermediate nodes $\mathbf{c}_{1:N}$:

$$\mathbf{c}_{1:N}^* = \operatorname{argmin}_{\mathbf{c}_{1:N}} \sum_{n=1}^{N+1} \|f(\mathbf{c}_n) - f(\mathbf{c}_{n-1})\|. \quad (8)$$

Question E.1: Is the piecewise-linear latent path with nodes $\{\mathbf{c}_0, \dots, \mathbf{c}_{N+1}\}$ a discrete approximation of the shortest path in the *latent space* $\mathbb{R}^d$, or the shortest path on the *manifold* $f(\mathbb{R}^d) \subset \mathbb{R}^D$?

- A ☐ It approximates the shortest path in the latent space $\mathbb{R}^d$.
- B ☐ It approximates the shortest path on the manifold $f(\mathbb{R}^d)$.

Explain your reasoning.

Question E.2: A student proposes to instead minimise $\sum_{n=1}^{N+1} \|\mathbf{c}_n - \mathbf{c}_{n-1}\|$ (straight lines in latent space). Would this produce the same result as eq. (8) in general?

- A ☐ Yes, they are always equivalent.
- B ☐ No, they differ in general.

Give a brief argument.

Exercise F — Curve energy

Consider the generative model $\mathbf{y} = f(\mathbf{x})$, where

$$f(\mathbf{x}) = \begin{bmatrix} x_1, & x_1^2 + x_2^2, & x_2 \end{bmatrix}^\top. \quad (9)$$

Consider the two latent curves $\mathbf{c}_1(t) = t[1, 0]^\top$ and $\mathbf{c}_2(t) = t[0, 1]^\top$ for $t \in [0, 1]$.

Question F.1: Derive the pull-back metric $G_{\mathbf{x}} = J_{\mathbf{x}}^\top J_{\mathbf{x}}$.

Question F.2: Evaluate $E(\mathbf{c}_1)$ and $E(\mathbf{c}_2)$ using $E(\mathbf{c}) = \int_0^1 \dot{\mathbf{c}}^\top G_{\mathbf{c}(t)} \dot{\mathbf{c}} \, dt$.

$$E(\mathbf{c}_1) =$$

$$E(\mathbf{c}_2) =$$

Sketch your derivations below.

Exercise G — Neural network reparametrization

Consider the neural network $f : \mathbb{R} \rightarrow \mathbb{R}$

$$f(x \mid w_1, w_2) = w_1 \sigma(w_2 x), \quad (10)$$

where $\sigma(x) = \max(0, x)$ is the ReLU activation function.

Question G.1: Let $w_3 = \alpha w_1$ for some $\alpha > 0$. Derive a value for w_4 such that

$$f(x \mid w_1, w_2) = f(x \mid \alpha w_1, w_4) \quad \text{for all } x. \quad (11)$$

$$w_4 =$$

Sketch your derivations below.

Question G.2: Discuss the implication of your answer for the statistical identifiability of the parameter vector (w_1, w_2) : can we uniquely determine (w_1, w_2) from observations $\{(x_i, f(x_i))\}$?

Question G.3: Would the same issue arise if σ were a *bijective* activation function (e.g., $\sigma(x) = x$)? Give a brief argument.

Part III Graph models

Exercise H — Shallow embeddings

We use the *sigmoid decoder* $P(A_{uv} = 1) = \sigma(\mathbf{z}_u^\top \mathbf{z}_v + b)$ where $\sigma(x) = 1/(1 + e^{-x})$, in a 2-dimensional embedding space with $\mathbf{z}_v = [x, y]^\top$.

Question H.1: Let $\mathbf{z}_u = [1, 0]^\top$ and $b = 0$. Where could node v be embedded so that $P(A_{uv} = 1) = 0.5$?

- A ☐ On the line $x = 0$ (the y -axis)
- B ☐ On the line $x = 1$
- C ☐ On a circle of radius 1 centred at the origin
- D ☐ On the line $y = 0$
- E ☐ None of the above

Show the main steps of your derivation.

Question H.2: Two nodes u and v are embedded as $\mathbf{z}_u = [2, 0]^\top$ and $\mathbf{z}_v = [0, 2]^\top$. With $b = 0$, which node pair has higher probability of an edge according to the sigmoid decoder?

- A ☐ Node pair (u, u) , i.e. a self-loop on u
- B ☐ Node pair (u, v)
- C ☐ Both pairs have equal probability
- D ☐ Cannot be determined without knowing b

Give a brief argument.

Exercise I — Graph Laplacians and random walks

Consider a graph whose random-walk Laplacian $L_{\text{rw}} = I - D^{-1}A$ is

$$L_{\text{rw}} = \begin{bmatrix} 1 & -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix},$$

with nodes labelled a, b, c, d in row order.

Question I.1: Sketch the graph.

Question I.2: A random walk starts at node b . What is the probability p of being at node a after exactly 2 steps?

$$p =$$

Give a brief argument.

Question I.3: The stationary distribution $\boldsymbol{\pi}$ of the random walk satisfies $D^{-1}A\boldsymbol{\pi} = \boldsymbol{\pi}$ (equivalently, $L_{\text{rw}}\boldsymbol{\pi} = \mathbf{0}$). Without full computation, write down the stationary distribution $\boldsymbol{\pi}$ up to normalisation. Recall that for an undirected graph, $\pi_v \propto d_v$.

$$\boldsymbol{\pi} \propto$$

Exercise J — Graph convolution

Consider the same 4-node star graph from Exercise I (centre a , leaves b, c, d). Its adjacency matrix and eigendecomposition are:

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \quad \lambda_1 = -\sqrt{3}, \lambda_2 = \lambda_3 = 0, \lambda_4 = \sqrt{3}.$$

A graph convolution is $\mathbf{y} = \sum_{k=0}^N h[k] A^k \mathbf{x}$.

Question J.1: Using the filter $h[0] = 1, h[1] = -1$ (and $h[k] = 0$ otherwise) and the signal $\mathbf{x} = [0, 1, 1, 1]^\top$, compute the filtered signal $\mathbf{y}$.

$$\mathbf{y} = \begin{bmatrix} & & & \end{bmatrix}^\top$$

Show the main steps. (*Hint:* Compute $A\mathbf{x}$ directly from the adjacency structure.)

Question J.2: The Fourier-domain filter matrix for filter $h[k]$ is $M = \sum_{k=0}^N h[k] \Lambda^k$. For $h[0] = 1, h[1] = -1$, what is the first diagonal entry M_{11} (corresponding to $\lambda_1 = -\sqrt{3}$)?

$$M_{11} =$$